giving, when the system is in thermal equilibrium,

$$\frac{\mathrm{d}}{\mathrm{dE_A}} \ln Z_\mathrm{A}(\mathrm{E_A}) = \frac{\mathrm{d}}{\mathrm{dE_B}} \ln Z_\mathrm{B}(\mathrm{E_B}) = \frac{1}{\mathrm{k_B T}} =: \beta. \tag{6.2}$$

We shall refer to the quantity β as the system's inverse or reciprocal temperature.

The volume of states of the active system with energy α is $Z_\mathrm{A}(\alpha)$. On the other hand the probability for the active system to have energy $\leq \alpha$ is the same as the probability of the bath having energy $\geq \mathrm{E} - \alpha$, this must be

$$P(\alpha) = \Pr\{\mathrm{E_B} \geq \mathrm{E} - \alpha\} \propto \int_0^\alpha Z_\mathrm{B}(\mathrm{E} - s)\mathrm{d}s.$$

Since $Z_\mathrm{B}(\mathrm{E} - s) = \exp(S_\mathrm{B}(\mathrm{E} - s)/\mathrm{k_B})$, and $s \ll \mathrm{E}$, we (naively[3]) apply the mean value theorem to S_B to obtain

$$S_\mathrm{B}(\mathrm{E} - s) = S_\mathrm{B}(\mathrm{E}) - S'_\mathrm{B}(\xi)s,$$

and from $S'_\mathrm{B}(\xi) = 1/\mathrm{T}$, we obtain

$$Z_\mathrm{B}(\mathrm{E} - s) \propto \exp(-\beta s),$$

where $\beta = (\mathrm{k_B T})^{-1}$. Thus we have

$$P(\alpha) \propto \int_0^\alpha \exp(-\beta s)\mathrm{d}s.$$

This is the probability of finding any given active system of our ensemble having energy $\leq \alpha$. The volume (or number, in the case of discrete systems) of states available at energy α is $Z_\mathrm{A}(\alpha)$. Thus the energy density of the canonical ensemble at energy level α is proportional to $Z_\mathrm{A}(\alpha)\exp(-\beta\alpha)$.

6.1.3 The Gibbs-Boltzmann Probability Density

Observe that the previous discussion does not quite tell us the form of the phase space density associated to the canonical ensemble. An elegant way to derive this

[3]This is one of several imprecisions in this discussion whose correction would entail significant additional labor; since our ultimate aim is to use the framework and not to provide its rigorous derivation, we prefer to retain the intuitive continuity of presentation, in keeping with popular treatments.